

lean4ml

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Theorem 1 (Smale 1958). *There is a homotopy of immersions of S^2 into $\mathbb{R}^3$ from the inclusion map to the antipodal map $a : q \mapsto -q$.*

Proof. This obviously follows from what we did so far. $\square$

TaylorAndSmoothness

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, $L \in \mathbb{R}_{\geq 0}$ (encoded in Lean as `NNReal`), and $s \subseteq E$.

Definition 2 (L -smoothness on a set). We say that f is L -smooth on s if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in s.$$

In Lean, this is encoded as `LipschitzOnWith L (fun x => gradient f x) s`.

Definition 3 (Quadratic upper model). The quadratic upper model on s is the property

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2, \quad \forall x, y \in s.$$

Theorem 4 (First-order Taylor formula along a line). *If f is C^1 , then for all $x, p \in E$,*

$$f(x + p) - f(x) = \int_0^1 \langle \nabla f(x + tp), p \rangle dt.$$

Proof. This is proved in Lean by first establishing a vector-valued fundamental theorem of calculus along the path $t \mapsto x + t \cdot p$, and then rewriting the Fréchet derivative in terms of the gradient. $\square$

Lemma 5 (Pointwise bound on the line-segment integrand). *Assume f is L -smooth on s , $x \in s$, and $x + \gamma(y - x) \in s$ with $\gamma \geq 0$. Then*

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x \rangle \leq L\gamma\|y - x\|^2.$$

Proof. In Lean, we first convert Lipschitz-on-set smoothness to a norm inequality for gradient differences, then apply Cauchy–Schwarz to bound the inner product by a product of norms, and finally rewrite the segment displacement norm as $\gamma\|y - x\|$ (for $\gamma \geq 0$). $\square$

Theorem 6 (Quadratic upper bound from smoothness on a convex set). *Assume f is C^1 , L -smooth on a convex set s , and $x, y \in s$. Then*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

Proof. The Lean proof follows exactly the standard decomposition:

1. apply Theorem 4 with $p := y - x$ to write $f(y) - f(x)$ as an integral of $\langle \nabla f(x + t(y - x)), y - x \rangle$;
2. subtract the constant term $\langle \nabla f(x), y - x \rangle$ inside the integral, rewriting the remainder as an integral of $\langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle$;
3. use convexity of s to ensure $x + t(y - x) \in s$ for all $t \in [0, 1]$;
4. bound the integrand pointwise using Lemma 5 (with $\gamma = t$) and integrate the bound on $[0, 1]$;
5. evaluate $\int_0^1 Lt dt = L/2$ to obtain the desired inequality.

$\square$

HessianVsSmoothness

Definition 7 (Hessian). The Hessian of f at x is the Fréchet derivative of the gradient map:

$$\nabla^2 f(x) := D(\nabla f)(x),$$

encoded in Lean as `fderiv` `(fun y => gradient f y) x`.

Theorem 8 (Hessian operator-norm bound $\Rightarrow$ global smoothness). *If the gradient map is differentiable everywhere and its derivative (the Hessian) has operator norm uniformly bounded by L , then f is L -smooth on $\mathbb{R}$ -universe (i.e. on `Set.univ`).*

Proof. In Lean, this is a direct application of a Mathlib theorem `lipschitzWith_of_nnnorm_fderiv_le`, upgraded to the `Set.univ` version via `lipschitzOnWith_univ`. $\square$

Theorem 9 (Global smoothness $\Rightarrow$ Hessian quadratic-form upper bound). *If f is L -smooth on `Set.univ`, then for all x, p ,*

$$\langle \nabla^2 f(x)p, p \rangle \leq L\|p\|^2.$$

Proof. The Lean proof extracts an operator-norm bound on $\nabla^2 f(x)$ from Lipschitz smoothness using the lemma `opNorm_fderiv_le_of_lipschitz`, and then bounds the quadratic form $\langle (\nabla^2 f(x))p, p \rangle$ by $\|\nabla^2 f(x)\| \|p\|^2$ via Cauchy–Schwarz and the operator norm inequality. $\square$

NecessaryConditions

Theorem 10 (First-order necessary condition). *If f is continuously differentiable and x is a local minimizer of f , then*

$$\nabla f(x) = 0.$$

Proof. In Lean, we use the Mathlib lemma `IsLocalMin.hasFDerivAt_eq_zero` applied to the Fréchet derivative of f at x , expressed via `HasGradientAt`. Converting the resulting identity in the dual space back to an inner-product statement yields $\langle \nabla f(x), \nabla f(x) \rangle = 0$ and hence $\nabla f(x) = 0$. $\square$

Theorem 11 (Second-order necessary condition: Hessian is PSD). *Suppose f is C^2 and x is a local minimizer of f . Then for all p ,*

$$0 \leq \langle \nabla^2 f(x)p, p \rangle.$$

Proof. The Lean proof is by contradiction. Assuming $\langle \nabla^2 f(x)p, p \rangle < 0$, one uses an exact second-order Taylor expansion with an integral remainder (proved in the file `NecessaryCondition.lean`) to show that for sufficiently small $\alpha > 0$,

$$f(x + \alpha p) - f(x) < 0,$$

contradicting local minimality. The analytic core is controlling the integral remainder using continuity of the Hessian along the line segment. $\square$

Sufficient Conditions

Definition 12 (Strict local minimizer). A strict local minimizer is a point x for which there exists a neighborhood U of x such that $f(x) < f(y)$ for all $y \in U$ with $y \neq x$.

Lemma 13 (Taylor lower bound from a uniform Hessian lower bound). Assume f is C^2 and $\nabla f(x) = 0$. If there exist $\varepsilon > 0$ and $\rho > 0$ such that for all $y \in B(x, \rho)$ and all p ,

$$\varepsilon \|p\|^2 \leq \langle \nabla^2 f(y) p, p \rangle,$$

then for all p with $\|p\| < \rho$,

$$\frac{\varepsilon}{2} \|p\|^2 \leq f(x + p) - f(x).$$

Proof. This is obtained by integrating the pointwise Hessian lower bound inside the exact double-integral remainder form of Taylor's theorem (proved earlier in the necessary-condition development), and then evaluating the resulting elementary integral on $[0, 1]$. $\square$

Theorem 14 (Second-order sufficient condition (uniform form)). Suppose f is C^2 and $\nabla f(x) = 0$. If there exist $\varepsilon > 0$ and $\rho > 0$ such that for all $y \in B(x, \rho)$ and all p ,

$$\varepsilon \|p\|^2 \leq \langle \nabla^2 f(y) p, p \rangle,$$

then x is a strict local minimizer of f .

Proof. Take the neighborhood $U := B(x, \rho)$. For any $y \in U$ with $y \neq x$, set $p := y - x$ so that $y = x + p$ and $\|p\| < \rho$. The previous lemma gives a strictly positive lower bound on $f(x + p) - f(x)$, hence $f(x) < f(y)$. $\square$

Theorem 15 (Second-order sufficient condition (positive definite Hessian, finite-dimensional)). In finite dimensions, if f is C^2 , $\nabla f(x) = 0$, and the Hessian at x is positive definite in the sense that

$$p \neq 0 \implies 0 < \langle \nabla^2 f(x) p, p \rangle,$$

then x is a strict local minimizer.

Proof. In Lean, the proof packages the analytic step “positive definite at x implies a uniform coercivity bound on a small ball around x ” (proved using compactness of the unit sphere in finite dimensions), and then applies the uniform-form sufficient condition. $\square$

GradientFlow

Definition 16 (Gradient flow certificate). A *gradient flow* for $f : E \rightarrow \mathbb{R}$ with initial point $x_0 \in E$ on a time interval $[0, T]$ is a trajectory $x : \mathbb{R} \rightarrow E$ such that $x(0) = x_0$ and $\dot{x}(t) = -\nabla f(x(t))$ for all $t \in [0, T]$.

Theorem 17 (Gradient flow existence on a finite time horizon). If f is C^1 , globally L -smooth, $T > 0$, and $x_0 \in E$, then there exists a gradient flow solution on $[0, T]$ starting at x_0 .

Proof. The Lean proof proceeds in three steps.

1. (*Uniform local existence*) Using Picard–Lindelöf for the ODE $\dot{x}(t) = -\nabla f(x(t))$, global L -smoothness gives a uniform step size $\delta = \frac{1}{4L+4}$ such that for every starting point x_0 there is a solution on $[0, \delta]$.

2. (*Iteration / extension*) By concatenating these local solutions, we obtain a solution on $[0, n\delta]$ for any $n \in \mathbb{N}$.
3. (*Restriction*) Taking n large enough so that $T \leq n\delta$ and restricting the trajectory yields a solution on $[0, T]$.

□

Definition 18 (Global gradient flow). A *global* gradient flow for $f : E \rightarrow \mathbb{R}$ with initial point x_0 is a trajectory $x : \mathbb{R} \rightarrow E$ such that $x(0) = x_0$ and $\dot{x}(t) = -\nabla f(x(t))$ for all $t \geq 0$.

Theorem 19 (Global gradient flow existence for all $t \geq 0$). *If f is C^1 and globally L -smooth, then for every $x_0 \in E$ there exists a global gradient flow solution defined for all $t \geq 0$.*

Proof. For each $n \in \mathbb{N}$, apply the finite-horizon existence theorem to obtain a solution on $[0, n + 1]$. Uniqueness of solutions on overlapping intervals implies these flows are consistent under restriction. Define the global trajectory by

$$x(t) := x_{\lceil t \rceil}(t),$$

i.e. evaluate the $(\lceil t \rceil)$ -th finite-horizon flow at time t . A locality (*eventually equal*) argument shows this global definition agrees with one fixed finite-horizon solution in a neighborhood of each $t \geq 0$, so differentiability and the ODE identity transfer pointwise, yielding a global gradient flow. □